PassFrame
tuition

Number Sequences & Patterns

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Find the next two terms: 4, 8, 12, 16, __, __.
Method & answer: Add 4 each time. Next terms: 20, 24.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Select ALL the sequences below that step up by 3 each time: A) 5,8,11,14 B) 2,4,6,8 C) 9,12,15,18 D) 3,6,9,12 E) 20,22,24,26

2. Find the next two terms: 6, 12, 18, 24, __, __.

3. Find the next two terms: 2, 5, 8, 11, __, __.

4. Find the missing term: 3, 6, __, 12, 15. A) 8 B) 9 C) 10 D) 11

5. Find the next two terms: 90, 80, 70, __, __.

6. Find the next two terms: 4, 8, 12, __, __.

7. Which is the rule for the sequence 7, 14, 21, 28…? A) add 5 B) add 7 C) add 9 D) multiply by 4

8. Find the missing term: 5, 10, __, 20, 25.

9. Find the next two terms: 60, 54, 48, __, __.

10. Select ALL the sequences below that step up by 8 each time: A) 8,16,24,32 B) 5,10,15,20 C) 3,11,19,27 D) 6,14,22,30 E) 9,18,27,36

11. Find the missing term: 12, 24, __, 48, 60.

12. Find the next two terms: 2, 6, 10, 14, __, __.

13. Which is the rule for the sequence 30, 27, 24, 21…? A) subtract 2 B) subtract 3 C) subtract 4 D) add 3

14. Find the next two terms: 9, 18, 27, __, __.

15. Find the missing term: 11, __, 19, 23, 27.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the next two terms: 5, 10, 20, 40, __, __.

2. Find the next two terms: 4, 9, 16, 25, __, __.

3. Find the next two terms: 6, 10, 15, 21, __, __.

6. The nth term of a sequence is 2n − 1. Find the 6th term.

7. Find the next two terms: 0.2, 0.4, 0.6, 0.8, __, __.

8. Select ALL the sequences below that decrease by the same amount each time: A) 25,21,17,13 B) 30,24,20,15 C) 50,40,30,20 D) 18,15,11,8 E) 90,80,70,60

11. The nth term of a sequence is 6n. Which of these is the 7th term? A) 36 B) 40 C) 42 D) 48

12. Find the next two terms: -8, -4, 0, 4, __, __.

13. Find the next term: 200, 100, 50, __. A) 20 B) 25 C) 30 D) 0

4. Select ALL the sequences below where each term doubles the one before: A) 4,8,16,32 B) 3,6,9,12 C) 5,10,20,40 D) 2,5,8,11 E) 6,12,24,48

5. The nth term of a sequence is 3n + 1. Which of these is the 3rd term? A) 9 B) 10 C) 11 D) 12

9. Find the missing term: 2, 6, 18, __, 162.

10. Find the next two terms: 2, 3, 5, 8, 13, __, __.

14. Find the rule and the next term: 3, 9, 27, 81, __.

15. Find the missing term: 5, 10, 20, __, 80.

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the next term: 2, 3, 5, 8, 12, __.

2. The nth term of a sequence is $n^2 - n$. Find the 6th term.

3. $3n + 5 = 41$. What is n? A) 10 B) 11 C) 12 D) 13

4. Select ALL of these that are square numbers: A) 16 B) 20 C) 25 D) 30 E) 36

5. A sequence has nth term $5n - 3$. Which term of the sequence equals 47? A) the 8th B) the 9th C) the 10th D) the 11th

6. A stack of chairs has 1 chair in the top row, and each row below has 2 more chairs than the row above. How many chairs are in the 6th row?

7. A tiler lays a pattern where row 1 has 3 tiles, row 2 has 6 tiles, row 3 has 9 tiles, and so on. How many tiles are in row 12?

8. A savings account starts with £50 and £15 is added every month. How much money is in the account after 8 months?

9. A ball is dropped and each bounce reaches half the height of the previous one. If the first bounce reaches 160cm, how high is the 4th bounce?

10. The number of seats in each row of a theatre follows the pattern 20, 23, 26, 29,... How many seats are in row 15? A) 56 B) 59 C) 62 D) 65

11. A sequence of squares is built from matchsticks: 1 square uses 4 sticks, 2 squares in a row use 7 sticks, 3 squares use 10 sticks. How many matchsticks are needed for 15 squares in a row?

12. A pattern of dots forms triangular numbers: 1, 3, 6, 10,... How many dots are needed for the 9th triangular arrangement?

13. Every year, a tree grows 3cm more than it did the previous year. In its first year it grew 5cm. How much did it grow in its 7th year?

14. A bacteria colony doubles every hour, starting with 50 bacteria. Select ALL of these that are correct population counts at some hour: A) 100 B) 150 C) 300 D) 400 E) 800

15. A sequence of numbers is generated by the rule 'double the previous number and subtract 1', starting at 3. What are the first five terms?